Introduction to Materials Property Prediction

Dane Morgan, Ryan Jacobs (University of Wisconsin - Madison, Wisconsin, USA)

Ben Afflerbach (Fort Lewis College, Colorado, USA)

Part of TMS Online Course: Artificial Intelligence and Machine Learning

Part of Module 3: Approaches to Materials Property Prediction with Machine Learning

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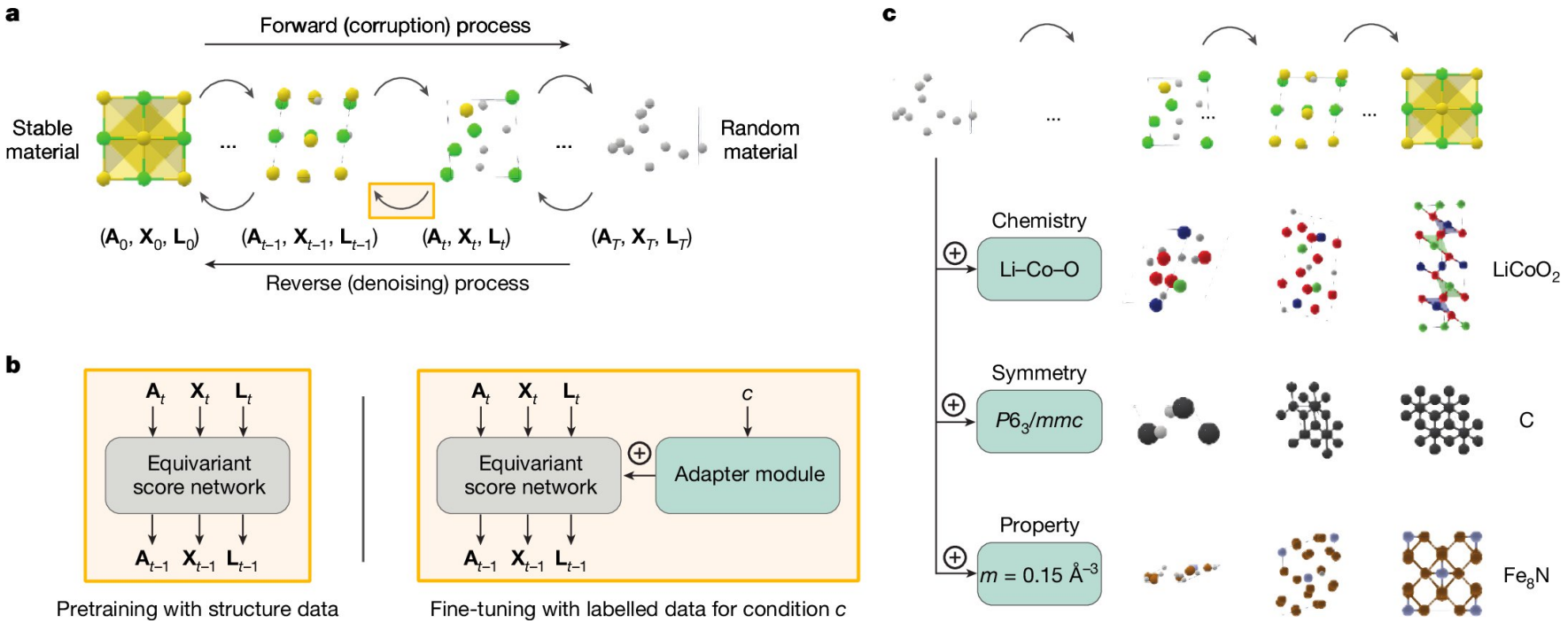
Moving Beyond Regression to Generative

- Generative machine learning on materials structures does 2 things
  - Generates a candidate materials structure
  - Predicts properties of that structure
- Generative step adds a whole additional machine learning challenge.
- Powerful for finding new materials with new properties

- Example of the MatterGen approach.
- Generates likely stable structures that also have target properties.
- Trained on large existing structural databases for structure generation and then can be fine-tuned for specific property. We will tune for Na-ion battery cathode voltages (Sec 3).



Zeni, C., et al. Nature 639 (2025). https://doi.org/10.1038/s41586-025-08628-5

Generating new materials with Mattergen

- Mattergen can be conditioned on target property values. E.g., new magnets with earth-abundant elements (low HHI score) and high magnetic density
- Conditioning Mattergen on both HHI score and magnetic density at the same time gave best results
- For this work, we will condition Mattergen on average Na voltage, chemical system, and energy above hull to target discovery of high voltage, stable materials in particular chemical systems, e.g,. Na-Mn-P-C-O

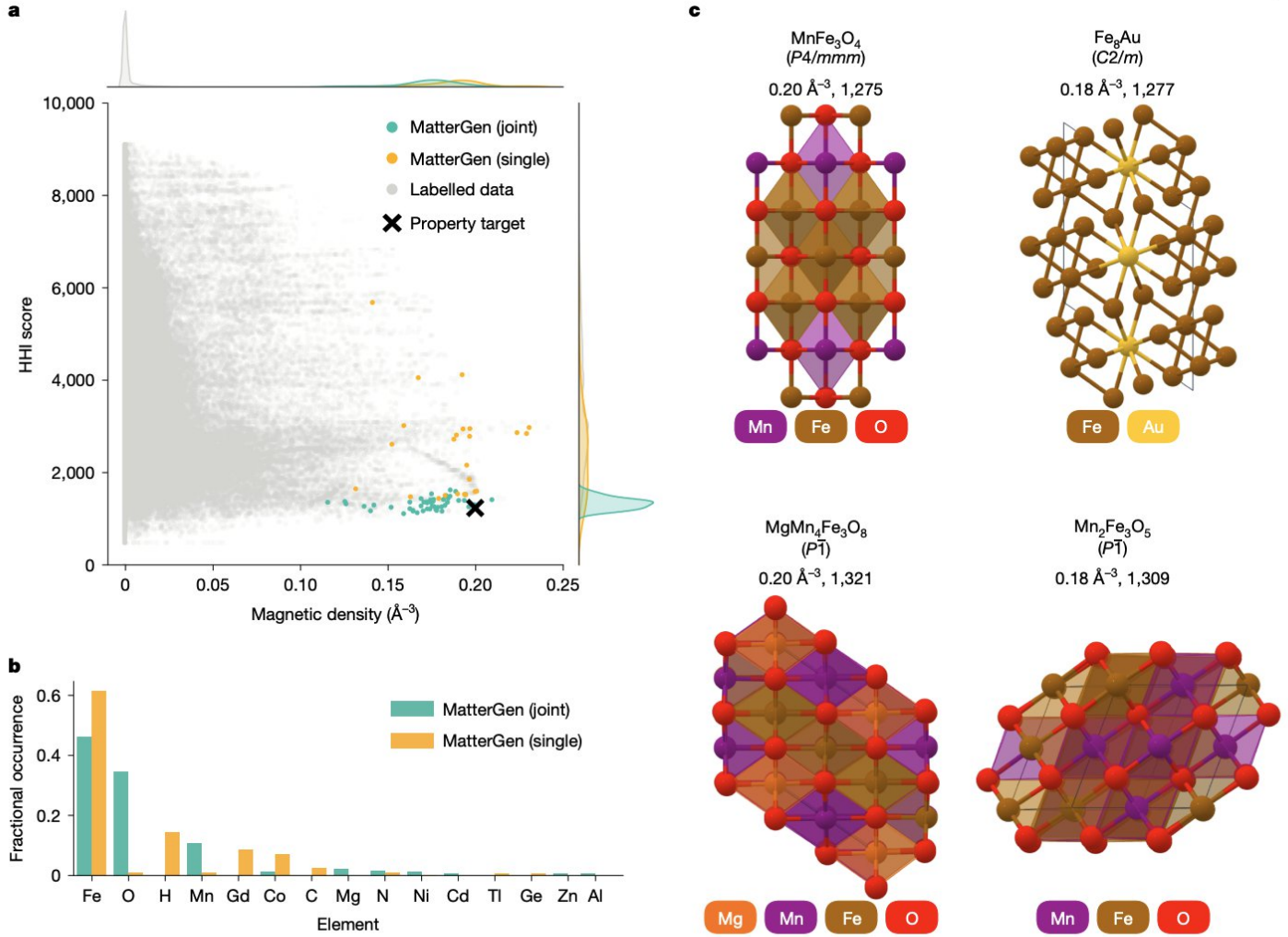


Fig. 5 | Designing low-supply-chain-risk magnets. a, Distribution of SUN structures generated by MatterGen when fine-tuned on magnetic density (single) and on both HHI score and magnetic density (joint), as well as structures from the labelled fine-tuning dataset. The property target of MatterGen is shown as a black cross. **b**, Occurrence of most frequent elements in SUN structures for the two fine-tuned MatterGen models. **c**, SUN structures on the Pareto front for the jointly fine-tuned model, along with their reduced formula, space group, magnetic density and HHI score.

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Mattergen Tutorial Notebook

- Our Mattergen tutorial will equip you with the tools needed to finetune Mattergen on your own data and for your own property of interest

- We will now move on to our lab activity, where we will do the following:
  - Fine-tune Mattergen using a database of UMLIP-calculated average voltages of prospective Na-cathode materials
  - Use our fine-tuned model to generate new potential materials in a prescribed composition space
  - Use a UMLIP (CHGNet) to calculate the energy of the discharged and charged state of a generated material and obtain its average voltage

TMS 2025 AI Workshop: Ryan Jacobs, Ben Afflerbach, Dane Morgan (UW-Madison)

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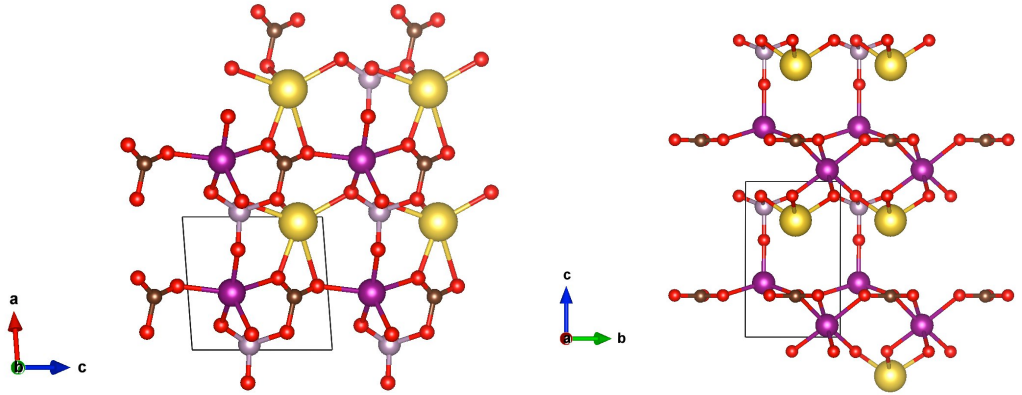
Google Colab notebook for finetuning Mattergen: example of conditional generation of high voltage Na-battery materials

MatterGen™